THE DEGREE MAP ON ZERO CYCLES FOR A CLASSICAL WEIL COHOMOLOGY THEORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file weil.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to closed points through compatibility of the cycle class map with pushforwards, and the computation of the degree of the class of a point.

Editorial note. The label lemma-degrees-cycles-classical names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source weil.tex –label lemma-degrees-cycles-classical –start 1694 –end 1716 –local-id 28172 returned RESULT=OK. Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The degree map of the spectrum of the base field. Axiom (C)(d) on the value of the degree map on the class of the base field. Provides the statement for the base field.
- Compatibility of the cycle class map with pushforwards. Axiom (C)(b) on the compatibility of the cycle class map with proper pushforwards. Expresses the class of a closed point as the pushforward of the class of the base field.

Source reference. Original section 0AZ0: sources/stacks-chow/tags.txt, source lines 13808–13808. Prerequisite pointer invoked at proof line(s) 1708; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0FGV: sources/stacks-weil/pinned-original-source.tex, source lines 1682–1682. Prerequisite pointer invoked at proof line(s) 1697; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:weil.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let H^* be a classical Weil cohomology theory (Definition 0FGV). Let X be a smooth projective variety of dimension d. The diagram*

$$\begin{array}{ccc} \mathrm{CH}^d(X) & \xrightarrow{\gamma} & H^{2d}(X) \\ \| & & \downarrow \int_X \\ \mathrm{CH}_0(X) & \xrightarrow{\deg} & F \end{array}$$

commutes where $\deg : \mathrm{CH}_0(X) \to \mathbf{Z}$ *is the degree of zero cycles discussed in Chow Homology, Section 0AZ0.*

Proof. The result holds for $\mathrm{Spec}(k)$ by axiom (C)(d). Let $x : \mathrm{Spec}(k) \to X$ be a closed point of X. Then we have $\gamma([x]) = x_*\gamma([\mathrm{Spec}(k)])$ in $H^{2d}(X)$ by axiom (C)(b). Hence $\int_X \gamma([x]) = 1$ by the definition of x_*. □